

## Experiment No. 4

### FRESNEL'S BIPRISM

#### I. OBJECTIVE

To determine the wavelength of sodium light using Fresnel's biprism method.

#### II. APPARATUS

Optical bench, sodium lamp, uprights, slit, biprism, convex lens, micrometer eyepiece.

#### III. THEORY

Fresnel's biprism experiment is an elegant method for studying the interference of light. A biprism essentially consists of two prisms (shaped from a single rectangular glass slab), each of very small refracting angle  $\alpha$  ( $0.5 - 1^\circ$ ) placed base to base (Figure-1). Monochromatic light of wavelength  $\lambda$  from a narrow slit  $S$ , adjusted parallel to the refracting edge  $B$ , is allowed to fall symmetrically on the biprism. The division of the incident wave front into two parts by the refracting edge takes place and the two emergent wave fronts appear to originate from virtual sources  $S_1$  and  $S_2$ , which being derived from the same source  $S$  act as coherent sources. As a result, interference fringes are observed in the overlapping region  $EF$  of the screen.

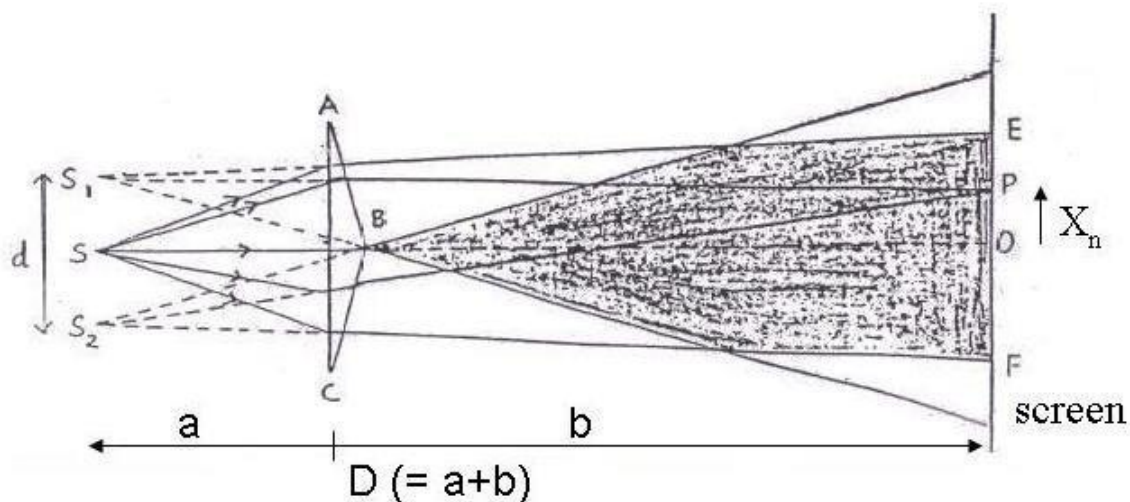


Figure 1: Division of wavefront by Fresnel's biprism

Let us consider a point P on the screen which receives two rays coming from the source S but refracting through the upper and lower parts of the biprism. As these rays appear to come from the virtual sources  $S_1$  and  $S_2$ , their path difference is given by  $(S_2P - S_1P)$

This path difference may be calculated as following:

Let  $S_1S_2 = d$  and  $SO = D$ . It can be easily be seen that

$$S_1P = D\left\{1 + \frac{1}{2}\left(\frac{\frac{d}{2} + X_n}{D}\right)^2\right\}$$

and

$$S_2P = D\left\{1 + \frac{1}{2}\left(\frac{\frac{d}{2} - X_n}{D}\right)^2\right\}$$

where,  $OP = X_n$  and assuming  $(\frac{d}{2} \pm X_n) \ll D$

Thus, the path difference is

$$S_2P - S_1P = \frac{d}{D}X_n$$

Now if  $X_n$  happens to be the position of the nth fringe, then ,

$$\frac{d}{D}X_n = n\lambda \quad (1)$$

and if the fringe width (x) is defined as the separation between neighboring bright (or dark) fringes, then,

$$x = X_n - X_{n-1} = \frac{\lambda D}{d} \text{ (from 1), i.e } \lambda = x \frac{d}{D} \quad (2)$$

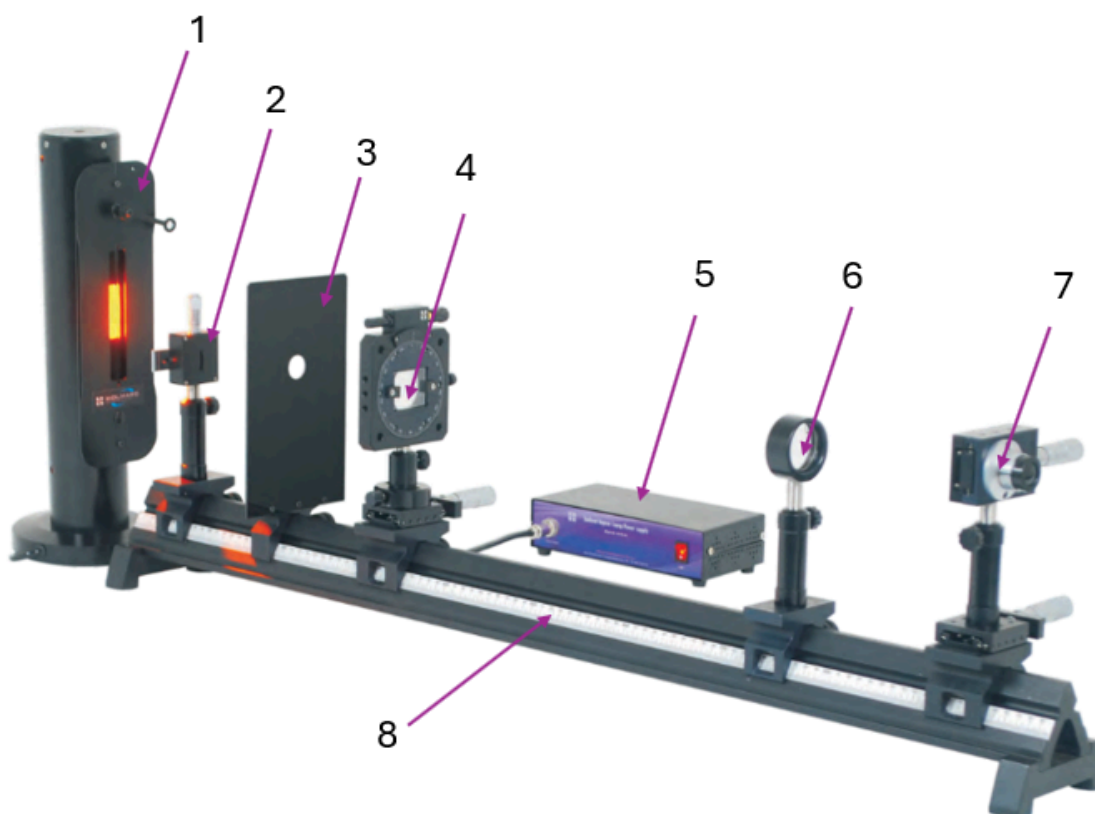
#### **IV. EXPERIMENTAL SETUP AND APPARATUS DETAILS**

The apparatus is as shown in Figure-2. The experiment is performed on an optical bench, which carries four uprights for holding the slit, biprism, convex lens and the micrometer eyepiece. Each of the upright can be moved along the length of the bench by pressing a knob at the base and can be locked in any position by releasing it. The height of the uprights can be changed by coarse adjustments. The upright for the biprism can also be moved perpendicular to the bench (laterally) with the help of a side knob on its base. The monochromatic light source used for the experiment is a sodium vapor lamp. A variable width slit is fixed on the upright closest to the light source and its position and height should not be normally disturbed. The next upright is used to hold the biprism. The biprism holder has a circular aperture. The holder can rotate the biprism in its own plane about the horizontal axis and can be locked. A finer adjustment is possible with a tangent screw in a locked position. The third upright is used for the lens holder which carries a convex

lens of given focal length. The last upright carries a micrometer eyepiece with a micrometer screw for lateral movement of crosswire in its field of view. The lateral position (X) of the vertical crosswire can be measured with the help of the micrometer screw and is given by,

$$X = M.S.R + C.S.R * L.C \quad (3)$$

M.S.R. C.S.R. and L.C. represent the main scale reading, circular scale reading and the least count of the micrometer screw respectively. Take into account the zero error if any.



**Figure 2: Apparatus for Fresnel's biprism experiment**

|    |                     |    |                                   |
|----|---------------------|----|-----------------------------------|
| 1. | Sodium lamp         | 5. | Power supply for sodium lamp      |
| 2. | Slit                | 6. | Convex Lens                       |
| 3. | Screen              | 7. | Eyepiece holder with a micrometer |
| 4. | Biprism with holder | 8. | Optical bench                     |

## V. PROCEDURE

1. Bring all the uprights close to the slit. Next, adjust their heights in such a way that the centers of the slit, the biprism and lens holders and the eyepiece are all at the same height. Also adjust their planes nearly perpendicular to the axis of the bench.
2. Mount the biprism on the holder with its plane face towards the slit. Position it such that the refracting edge of the biprism is at the center of the circular aperture. **With the help of side knob move the upright laterally so that the edge of the biprism is nearly in line with the slit.** Rotate the biprism holder in its plane to ensure that the edge of the biprism is nearly vertical, and then lock it. Now, the upright holding the biprism may be positioned at a distance of around 10 cm from the slit. Adjust the slit width to 'just open' condition.
3. Move the eyepiece upright to about a meter from the biprism. Remove the eyepiece and through the aperture, directly look towards the biprisms with naked eye. You should be able to see two virtual images of the slit. If not, slightly move the biprism laterally using the base knob on the upright.
4. Replace the eyepiece in its position. You should be able to see alternate dark and bright fringes on the screen right away. If not, move the biprism upright laterally (using side knob) to locate the fringes. Adjust the sharpness and contrast of the fringes by slightly rotating the biprism in its plane using the tangent screw. Note that the slit width may also have to be slightly adjusted to improve the sharpness of the fringes. The distance between the slit and the biprism determines the no. of fringes and fringe width. Adjust it within 515mm to get a reasonably large no. of fringes with appreciable width.
5. Next, move the eyepiece towards the biprism. If the fringes are found to move laterally with respect to the vertical cross wire, it implies that refracting edge of the biprism is away from the bench axis (Figure-3a), i.e. the horizontal line joining the slit and the edge of the biprism does not coincide with the bench axis as shown in Figure-3b. In a situation, as shown in Figure-3(a), the fringes are formed on a plane which is not normal to the bench axis, while the eyepiece is constrained to move along the bench axis. As a result, when the eyepiece is moved backward (away from biprism) or forward (towards biprism) the vertical crosswire appears to move across the fringe pattern. Notice that if the fringe width is measured in this situation, by moving the crosswire in a plane perpendicular to the bench axis, one would be measuring  $x/\cos \theta$
6. instead of the actual fringe width  $x$  (right side, Figure-3). If the initial adjustments are not done properly, the lateral shift may be so large that as the eyepiece is moved along the bench, the fringe system disappears from the field of view.

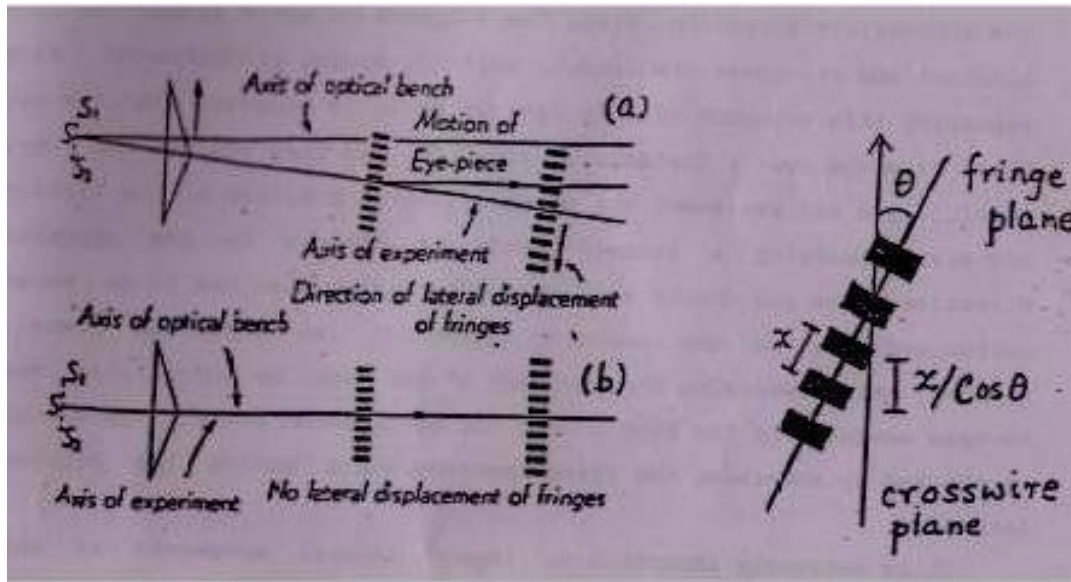


Figure 3: Lateral shift in Fresnel's biprism experiment

### REMOVAL OF LATERAL SHIFT

The following procedure may be adopted to remove lateral shift and make the fringe pattern stationary. First bring the eyepiece upright close to the biprism to see the fringes in the centre of its field of view. Let the vertical crosswire coincide with any bright fringe in the center of the fringe pattern. Now, move the eyepiece backwards and during this motion observe the fringe pattern. If the fringe pattern appears to shift left w.r.t. the vertical crosswire, then move the biprism laterally to the right, by slightly turning the side knob. **This can be checked by viewing the fringe through eye piece while moving the biprism laterally.** Alternatively, if the fringe pattern shifts to the right w.r.t. the crosswire, then biprism is to be moved laterally to the left. After moving the biprism laterally in the appropriate direction, bring the eyepiece up again (close to the biprism) and reinspect the lateral shift by moving it backwards. Keep repeating this sequence till no lateral shift is observed when the eye piece is moved over a distance of about 1 m. In case you are not very careful, you may overshoot the optimum lateral position of the biprism and start observing a lateral shift of fringes in the opposite direction. Now you should know that the biprism also has to be moved in the opposite direction (as explained above). However, while reversing the movement of the knob, be aware that the reverse movement of the biprism may not occur immediately. This can be checked by observing the fringe pattern while moving the biprism laterally.

It is extremely important to impart lateral movements to the biprism systematically and only in small steps followed by an inspection of lateral shift each time, by moving the eyepiece backwards. It must be realized that by virtue of the initial alignment of the setup

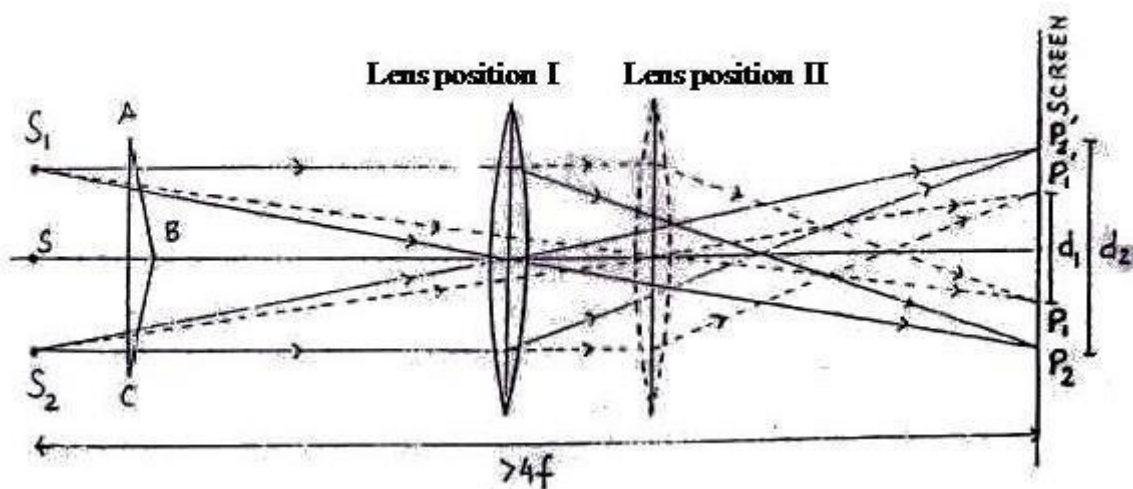
(steps 1-4), the biprism is quite close to the optimum position. Random and large lateral movements of the biprism or any other components make you completely lose this initial (default) advantage. Also note that during lateral shift removal, the eyepiece may also have to be sometimes moved laterally to maintain the fringe system in the centre of its field of view. Remember that any lateral movement of the eyepiece has no role in correcting the lateral shift of the fringes.

6. Once the lateral shift has been removed, make sure that the slit and the biprism are not disturbed in any way and both are locked at respective positions. However, before taking measurements, inspect the uprights and measure their positions on the optical bench. Carefully examine the main scale and the circular scale of the micrometer eyepiece. Record the value of each division on the main scale and the least count (L.C.) of the circular scale. While taking measurements, you must separately tabulate the main scale reading (M.S.R.) and the circular scale reading (C.S.R.) and then use the relation (3) (or an equivalent one) to obtain the position of the vertical crosswire.
7. Lock the eyepiece at a distance of about 1m from the slit and use the base knob of the eyepiece upright to make sure that the fringe system is at the centre of its field of view. Focus the crosswires and move the vertical crosswire to one end of the fringe system. Align the vertical crosswire parallel to a bright fringe, at the centre of its width (call it the zeroth fringe). Note down its position as  $x_0$  on the micrometer scale recording M.S.R. and C.S.R. separately. Next, turn the micrometer screw and move the vertical crosswire to the centre of some other bright fringe (say  $p$ th, with  $p = 2-5$ ) and note down its position ( $x_p$ ).

Keep moving the crosswire successively to other fringes (say 2 $p$ th, 3 $p$ th,.... And so on) and note down the respective positions. The value of  $p$  must be such chosen that you are able to take about 7 readings spread over the complete range of the fringe pattern. While taking measurements, the crosswire must be moved only in one direction to avoid error due to backlash. Even, in case you overshoot the position of a certain fringe, never move the crosswire backwards. Instead move to the just next bright fringe, measure its position and assign its correct fringe no. (w.r.t. the zeroth fringe).

8. Take down the positions of the slit and the eyepiece on the bench. The difference between these gives  $D$ . However, this value of  $D$  is to be corrected for bench error (as explained later).
9. **Measurement of distance ( $d$ ) between virtual sources:** Do not disturb the positions of the slit and biprism till the distance “ $d$ ” between the virtual sources is measured using the conjugate foci method (Figure-4). For this, a convex lens is mounted on the lens holder upright between the biprism and the eyepiece. If the eyepiece is placed at a distance from the slit, which is greater than four times the focal length ( $f$ ) of the lens and the lens is moved in between, then at two positions of the lens, magnified and diminished images of

the virtual sources are formed in the plane of the crosswires. If  $d_1$  and  $d_2$  are the separations between the magnified and diminished images of the virtual sources then the actual distance 'd' between virtual sources is given by  $\sqrt{d_1 d_2}$ . The distance between the eyepiece and the slit should be kept just greater than  $4f$ , so that  $d_1$  and  $d_2$  are not very different, and the error in the measurement of  $d$  is minimized.



**Figure 4: Conjugate foci method to determine d**

Holding the eyepiece in a particular position, move the lens holder to observe these two images and adjust the position of the eyepiece so that  $d_1$  and  $d_2$  are not very different in size. Lock the eyepiece in an optimum position and adjust the position of the lens to get sharp images of the virtual sources on the crosswire. Use the micrometer screw to note down the positions  $P_1$  and  $P_1'$  of the images of the two virtual sources in the field of view of the eye piece. Next adjust the position of the lens for the other pair of sharp images to be informed on the field of view of the eyepiece and note down the positions  $P_2$  and  $P_2'$  of the images of the virtual sources. Take  $d_1 = |P_1 - P_1'|$  and  $d_2 = |P_2 - P_2'|$  and  $d = \sqrt{d_1 d_2}$

Repeat the above procedure once again by repositioning the eyepiece.

## **VI. RESULTS AND CALCULATIONS**

1. Tabulate the fringe no. (n) and the fringe position  $X_n$  (M.S.R., C.S.R. and the total separately).

2. Tabulate  $(P_1, P_1', d_1)$ ,  $(P_2, P_2', d_2)$  and  $d = \sqrt{d_1 d_2}$ . Calculate the average value of  $d$  and  $\Delta d$  (which is half the magnitude of difference between the two measurements). Take  $\frac{\Delta d}{d_{avg}}$  as the fractional error in  $d$ .
3. Record the positions of the slit and the eyepiece on the optical bench. Their difference is the measured value of  $D$ . However, this value of  $D$  may not be the same as the actual distance between the slit and the plane of the crosswires. The difference  $[D(actual) - D(measured)]$  is called the bench error ( $\Delta$ ). In the present setup, the bench error is zero.
4. Plot a linear graph for  $X_n$  versus  $N$  and measure its slope. Calculate the average fringe width  $x = \frac{X_n}{n}$  from this slope. Also draw the limiting straight lines and measure their slopes ( $x_1$  and  $x_2$ ) to get the limiting values of  $x$ . Take the fractional error in  $x$  is  $\frac{\Delta x}{x} = \frac{|x_1 - x_2|}{2x}$ .
5. Calculate  $(\lambda = \frac{xd}{D})$  and estimate the error  $\Delta \lambda$  using the fractional errors  $\frac{\Delta d}{d}$  and  $\frac{\Delta x}{x}$  (see Section in Errors). The error in  $D$  may be neglected.

## **VII. PRECAUTIONS**

1. Distance between the slit and the biprism must be kept constant throughout the experiment. After the initial adjustment to get sharp fringes with good contrast, do not change the slit width or rotate the biprism.
2. Move the eyepiece only backwards to observe the lateral shift. For moving lateral shift, there is no other adjustment required except the lateral movement of the biprism.
3. During measurement of  $d_1$  and  $d_2$ , don't keep the eyepiece at a very large distance ( $\gg 4f$ ) from the slit, which results in a large difference in  $d_1$  and  $d_2$  leading to enhanced error in their product.
4. While measuring the fringe width, align the crosswire at the centre of a bright fringe.
5. While measuring the fringe width, the pointer (crosswire) must be moved only in one direction to avoid error due to backlash.

## **VIII. LEARNING OUTCOMES**

To study interference of light due to division of the wavefront using Fresnel's biprism.

## **IX. REFERENCE**

B.K. Mathur: Introduction to geometrical and Physical optics.